Namal University, Mianwali
Department of Electrical Engineering

Course Title: EE-253 Data Structures and Algorithms

Lab Project Report

Title: Library Management System

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Instructions

1. This is an individual lab. You will perform the tasks individually and submit a report.
2. Some of these tasks (marked as 'Example') are for practice purposes only while others (marked as 'Task') have to be answered in the report.
3. When asked to display an output in the task, either save it as jpeg or take a screenshot, in order to insert it in the report.
4. The report should be submitted on the given template, including:
    a. Code (copy and pasted, NOT a screenshot)
    b. Output figure (as instructed in 3)
    c. Explanation where required
5. The report should be properly formatted, with easy to read code and easy to see figures.
6. Plagiarism or any hint thereof will be dealt with strictly. Any incident where plagiarism is caught, both (or all) students involved will be given zero marks, regardless of who copied whom. Multiple such incidents will result in disciplinary action being taken.
7. Late submission of report is allowed within 03 days after lab with 20% deduction of marks every day.
8. You have to submit report in pdf format (Reg.X_DSA_LabReportX.pdf).

Project Statement:

Design a program that can manage a library's catalog system. The program should allow library staff to add, remove, and modify books in the library's collection. The program should also allow users to search for books based on title, author, or subject.

Program Requirement Features:

The program should have the following features:

1. The books in the library's collection should be stored in an array, and information about each book, such as the title, author, and subject, should be associated with it.

2. The program should include a search algorithm that allows users to search for books based on title, author, or subject. The program should display a list of matching books and their availability status.

3. The program should allow staff to add, remove, or modify books in the library's collection. When a book is added or removed, the program should update the array accordingly.

4. The program should use a sorting algorithm to sort the books in the library's collection based on title or author.

5. The program should have a graphical user-friendly interface that is easy to navigate and use

INTRODUCTION:

The Library Management Program is designed to manage a library catalog and provide various functionalities for library staff, and students.

Program Features Explanation:

❖ Add Book:

This functionality will add a new book to the library catalog with the book title, author name and description or subject of the book.

```
*******************************************************************************
             Welcome to Namal Library Management System
*******************************************************************************
Login as:
1:- Librarain
2:- Student
3:- Close Program
Enter your choice (1 to 3): 1
librarian Login
Enter your username: namal
Enter your password: 123
Library Management System
1:- Add Book
2:- Remove Book
3:- Search Book
4:- Modify Book
5:- Display Books
6:- Exit
Enter your choice (1-6): 1
Enter the book name: Pakistan
Enter the book author: Imran Khan Niazi
Enter the book subject: Personal history
Book added successfully!
```

❖ Remove Book:

This functionality will Remove a book from the library catalog. Also the details associated with the book like book author and subject of the book will also be removed.

```
Library Management System
1:- Add Book
2:- Remove Book
3:- Search Book
4:- Modify Book
5:- Display Books
6:- Exit
Enter your choice (1-6): 2
Enter the book name: Pakistan
Book removed successfully!
Title: Pakistan
 Author: Imran Khan Niazi
 Subject: Personal history
```

❖ Modify Book:

Modify the details of a book in the library catalog. By using this functionality, user will be able to change the title of the book.

```
Library Management System
1:- Add Book
2:- Remove Book
3:- Search Book
4:- Modify Book
5:- Display Books
6:- Exit
Enter your choice (1-6): 4
Enter the book name to be modified: Pakistan
Enter the new book name: The unspoken words
```

- Search Books:

Search for books in the library catalog based on title and the details associated with the book like book author and subject of the book will also be displayed.

```
Library Management System
1:- Add Book
2:- Remove Book
3:- Search Book
4:- Modify Book
5:- Display Books
6:- Exit
Enter your choice (1-6): 3
Enter the book name: CP
Book found in the library.
Title: CP
author: Sir Bacha
subject: Computer Programming
```

- Display Books:

This functionality will display all books available in the library catalog. Also the details associated with the book like
book author and subject of the book will also be displayed on the screen.

```
Books available in the library:

Title: BCA
author: Sajjad-ur-Rehman
subject: Basic Circuit
**********

Title: QCR
author: Mam Zulikha
subject: Quational Reasoning
**********

Title: FE
author: Sir Asif
subject: Functional English
**********

Title: QS-1
author: Sir Umer
subject: Quranic Sudies
**********

Title: ENA
author: Sir Usama
subject: Electric Network
**********

Title: CP
author: Sir Bacha
subject: Computer Programming
**********
```

- ❖ Existing the program:

This functionality will terminate the program and the user will not be able to visit the library anymore.

```
*****************************************************************************
            Welcome to Namal Library Management System
*****************************************************************************
Login as:
1:- Librarain
2:- Student
3:- Close Program
Enter your choice (1 to 3): 3
Exiting the program...
Thank You for visiting Namal Library!
```

Conclusion and Learning from the project:

Basically we learnt practical implementation of python. The knowledge and practice that we do in data structure course and lab was utilized in the making of this project. This project is able to run and stimulate at lower level successfully as there was no error. It can help mini libraries to manage their library catalog. This program can add a book, remove a book, search books, modify and display the books available in the library. We used file handling concept for the making of this project that was a new thing for us as we did it for the first time. Overall this project improves our python skills to make smaller program work successfully.

Lab Evaluation Rubrics							
Domain	**CLOs/ Rubric**	**Performance Indicator**	**Unsatisfactory 0-2**	**Marginal 3-5**	**Satisfactory 6-8**	**Exemplary 9-10**	**Allocate d Marks**
Psychomotor	**CLO:1 R2**	Implementation with Results **(P)**	Does not try to solve problems. Many mistakes in code and difficult to comprehend for the instructor. There is not result of the problem.	Does not suggests or refine solutions but is willing to try out solutions suggested by others. Few mistakes in code, but done along with comments, and easy to comprehend for the instructor. Few mistake in result.	Refines solutions suggested by others. Complete and error-free code is done. No comments in the code, but easy to comprehend for the instructor. Results are correctly produced.	Actively looks for and suggests solution to problems. Complete and error free code is done, easy to comprehend for the instructor. Results are correctly produced. Student incorporated comments in the code.	
Affective	**CLO:3 R3**	Lab Report **(A)**	Code of the problem is not given. Outputs are not provided. Explanation of the solution is not stated.	Code of the problem is given. Output is not complete. Explanation of the solution is not satisfactory.	Code of the problem is given. Output is completely given. Explanation of the solution is not satisfactory.	Code of the problem is given. Output is completely given. Explanation of the solution is satisfactory.	
	CLO:1 R5	Discipline and Behavior **(A)**	Got and wandered around. Chased others, ran, or played around. More than two incidents of talking non-lab related stuff in lab and/or any talk with other groups, voice level exceeding the appropriate level, use of cell phones and involvement in any non lab activity.	Got out of seat and wander around for some time. No more than two incidents of talking non-lab related stuff in lab. Voice level exceeding the appropriate level, use of cell phones and involvement in any non-lab related activity.	Stayed in seat and got up for a specific lab related reason, but took more time than required to do the job. No more than one incidents of talking non-lab related stuff in lab. Voice level exceeding the appropriate level, use of cell phones and involvement in any non-lab related activity.	Stayed in seat and got up for a specific lab related reason. Took care of lab related business and sat down right away. Voice level kept appropriate. Not used cell phones or involved in any non- lab related activity.	